

Day 1 — Course intro & what is R?

Monday, May 18, 2026

Today

Before class

- Read the syllabus (linked from Canvas).
- (Optional): Read *Data Computing* §1.

Plan for today

- Course logistics
- Introduction to R and RStudio
- Tidy data
- R Demo

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [Data Computing §1](#)

1. Course Info & Introductions

Summer Session 1: May 18, 2026 – June 26, 2026

- **Class time:** M/T/W/Th/F, 9:35 a.m. to 10:50 a.m.
- **Class location:** Huck Life Sciences Building 007
- **Instructor:** Charles Costanzo
 - Email: cgc5478@psu.edu
 - Office Hours: M/W/Th/F: 10:50-11:50 a.m. in Huck 005 (or on Zoom by appt)
- **Teaching Assistant (TA):** Xinyue Wang

– Email: xpw5228@psu.edu

💡 Icebreaker: name, major, your favorite song right now.

Learning Goals

This course will teach you how to:

- **Use R:** Program in the R *language* effectively and use the R *software environment* to efficiently compute with data.
- **Wrangle Real Data:** Access, clean, join, and format data from various sources.
- **Visualize Information:** Create graphical and descriptive summaries of data.
- **Build Reproducible Workflows:** Write well-documented code using tools like RStudio, RMarkdown, and Git/GitHub.

💡 No previous programming background is required for this course.

❗ This is **not** a course on general purpose programming.

Course logistics

- **Six weeks is short.** Daily pace is faster than a 15-week course. Falling a week—or even a day—behind is more serious than spring/fall courses.
- **Attendance matters.** Lots of in-class graded work (activities, quizzes, oral checkpoints).
- **Timely communication is critical.** Reach out to me **as soon as possible** if something comes up that necessitates you missing class, having a deadline extended, etc.
- **Assignment Weights:**

Readings	Quizzes	In-class activities	Homework	Exam	Course project
10%	15%	25%	10%	20%	20%

- **Grading Scheme:**

A	A-	B+	B	B-	C+	C	D	F
93%+	90%	87%	83%	80%	77%	70%	60%	< 60%

AI policy

- **Open assessments (Homework & Project):** AI tools are permitted **with mandatory disclosure**.
- **Closed-book, in-person work (Quizzes, Exam, Activities, Oral Checkpoints):** AI tools are **strictly prohibited**.
- **Disclosure is graded (1-2 points).** Omitting the disclosure loses points; lying on the disclosure is an academic integrity violation.
- **You own the output.** You must be able to explain any code you submit, which I will verify during in-class oral project checkpoints.
- **Practical implication:** AI is a study aid, not a substitute. Roughly 60% of your grade comes from verified in-person work where AI cannot be used.

Course Readings (Perusall)

- All assigned textbook readings are hosted on **Perusall**, a collaborative e-reading platform.
- **How grading works:** You earn credit through holistic engagement (reading the entire text, spending active time, leaving high-quality comments, and replying to classmates).
- **No access code needed!** You must access readings *directly through Canvas links* to automatically authenticate your Penn State account.

! Let's try it together right now:

1. Log into our Canvas course.
2. Click the "Perusall" tab on the left.
3. Click the link for **DC Section 1 (§1.1–1.5)**.

What is R?

R is a **programming language** *and* a **software environment** for statistical computing and graphics.

- Designed by statisticians for the purpose of data analysis.
- First version created in August 1993 (almost 33 years ago).

- Has grown rapidly in popularity and is widely used in academia and industry by statisticians, data analysts, and data scientists alike.
- Enormous number of user-contributed **packages**, which are add-ons that extend the functionality of R.
 - Currently the Comprehensive R Archive Network (CRAN) package repository features 23,670 available packages.¹

Some Examples of Said R Packages



Figure 1: Hex stickers from the [rstudio/hex-stickers](https://rstudio.github.io/hex-stickers/) repository.

R vs. RStudio

- **R** is a computer language that performs computations based on human-written instructions.
- **RStudio** is an integrated development environment (**IDE**) for R programming. It combines the R console with supplementary tools for more efficient R programming.

¹As of May 15, 2026.



(a) **R**: Do not open this



(b) **RStudio**: Open this

Figure 2: R and RStudio logos.

Tidy Data $\neq$ Neat Data

💡 This table is easy for humans to read. What might make it tricky for a computer to read?

COUNTY	Week Of 05-04-2026 to 05-11-2026				Yearly Total 2026			
	To Democratic		To Republican		To Democratic		To Republican	
	R	OTH	D	OTH	R	OTH	D	OTH
ADAMS	5	4	4	3	71	93	74	102
ALLEGHENY	47	125	23	31	766	1,323	810	389
ARMSTRONG	2	0	1	1	23	28	65	26
BEAVER	2	15	3	1	77	127	150	55
BEDFORD	0	1	0	1	13	14	21	16
BERKS	15	37	14	16	337	494	422	335
BLAIR	5	7	0	3	60	83	62	52
BRADFORD	1	2	1	2	32	27	40	45
BUCKS	32	94	15	20	626	1,054	550	557
BUTLER	3	5	0	3	131	170	101	111
CAMBRIA	1	2	3	0	82	71	174	62
CAMERON	0	0	0	0	1	1	7	0
CARBON	5	11	4	5	34	70	54	43
CENTRE	2	14	3	4	88	130	48	75
PHILADELPHIA	60	313	66	29	1,076	3,163	1,173	489
VENANGO	2	2	0	0	28	35	30	27
WARREN	0	0	1	1	24	32	15	24
WASHINGTON	5	7	7	3	112	142	188	111
WAYNE	1	1	3	2	22	35	41	39
WESTMORELAND	8	20	12	8	211	238	302	168
WYOMING	0	2	3	0	22	22	27	17
YORK	24	60	10	22	415	656	325	393
Totals:	512	1,348	334	322	9,823	15,604	9,631	7,295

Figure 3: A spreadsheet showing party change applications in PA. Source: PA Dept. of State (May 4–11, 2026).

Two simple rules for Tidy Data:

1. **Consistent Cases (Rows)**: Every row represents the same underlying attribute
2. **Uniform Variables (Columns)**: Every column contains the same type of value.

PA Voter Table - Tidied Up

```
# 1. Load data and provide clean column names to handle the nested headers
raw_data <- readxl::read_excel("data/currentvotestats.xlsx",
                              sheet = "Party-to-Party(2026)",
                              skip = 3, # Skip the human-readable headers
                              col_names = c("county",
                                             "week_dem_R", "week_dem_OTH",
                                             "week_rep_D", "week_rep_OTH",
                                             "year_dem_R", "year_dem_OTH",
                                             "year_rep_D", "year_rep_OTH"))

# 2. Clean and Pivot
tidy_voter_data <- raw_data %>%
  filter(!is.na(county), county != "Totals:") %>% # Rule 1: Consistent Cases
  pivot_longer(
    cols = -county,
    names_to = c("period", "target_party", "origin_party"),
    names_sep = "_",
    values_to = "count"
  )

knitr::kable(head(tidy_voter_data %>% filter(county == "CENTRE")))
```

county	period	target_party	origin_party	count
CENTRE	week	dem	R	2
CENTRE	week	dem	OTH	14
CENTRE	week	rep	D	3
CENTRE	week	rep	OTH	4
CENTRE	year	dem	R	88
CENTRE	year	dem	OTH	130

Demo: what R can do

```
library(tidyverse)
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

```
penguins, penguins_raw
```

```
# What does the data look like?
glimpse(penguins)
```

```

Rows: 344
Columns: 8
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex          <fct> male, female, female, NA, female, male, female, male~
$ year         <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~

```

```
# Quick summary by species
penguins |>
  group_by(species) |>
  summarize(
    n = n(),
    mean_mass_g = mean(body_mass_g, na.rm = TRUE),
    mean_bill_mm = mean(bill_length_mm, na.rm = TRUE)
  )
```

```
# A tibble: 3 x 4
  species      n mean_mass_g mean_bill_mm
<fct>    <int>    <dbl>      <dbl>
1 Adelie   152    3701.        38.8
2 Chinstrap  68    3733.        48.8
3 Gentoo   124    5076.        47.5
```

```
# A nice plot
penguins |>
  ggplot(aes(x = bill_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Bill length vs body mass, by species",

```

```

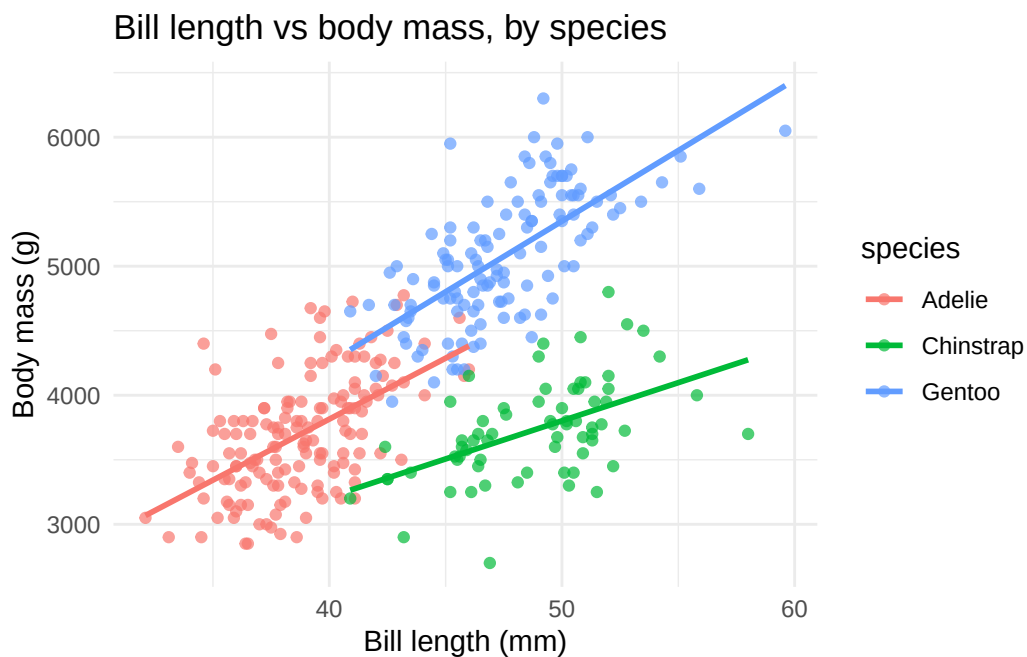
  x = "Bill length (mm)",
  y = "Body mass (g)"
) +
theme_minimal()

```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```

# Load necessary libraries
library(tidyverse)
library(sf)
library(tigris)
library(ggspatial)
library(patchwork)

# 1. Get the boundary for Centre County, PA

```



```

options(tigris_use_cache = TRUE)
centre_county <- counties(state = "PA", cb = TRUE, progress_bar = FALSE) |>
  filter(NAME == "Centre")

# 2. Define the exact coordinates for the Huck Life Sciences Building
huck_building <- tibble(
  lon = -77.86146,
  lat = 40.80100,
  name = "Huck Life Sciences"
) |>
  st_as_sf(coords = c("lon", "lat"), crs = 4326)

# 3. Create the Main Map (Centre County overview)
map_main <- ggplot() +
  geom_sf(data = centre_county, fill = "gray90", color = "gray50",
    linewidth = 0.5) +
  geom_sf(data = huck_building, color = "red", size = 3) +
  theme_void() +
  labs(title = "Centre County, Pennsylvania")

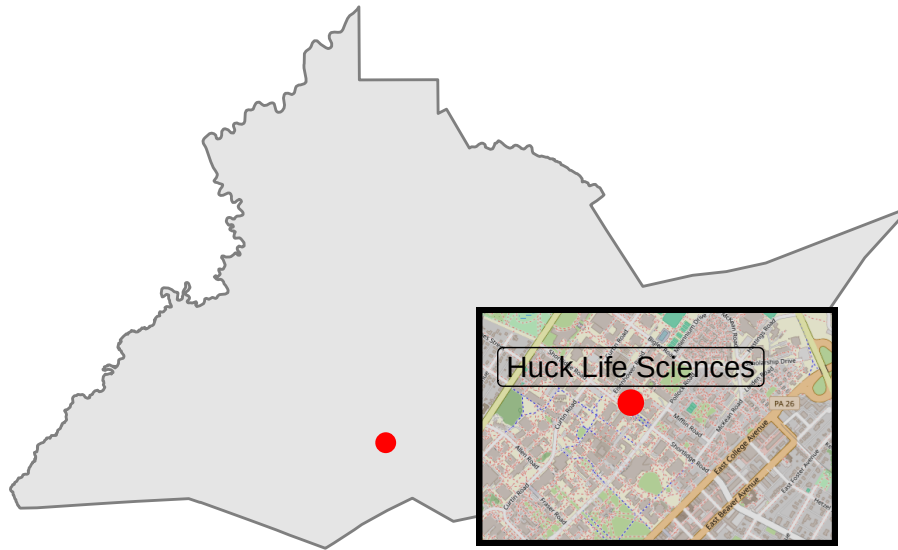
# 4. Create the Inset Map (Zoomed in on campus)
map_inset <- ggplot() +
  annotation_map_tile(type = "osm", zoom = 15, progress = "none") +
  geom_sf(data = huck_building, color = "red", size = 4) +
  geom_sf_label(data = huck_building, aes(label = name), nudge_y = 0.0015) +
  coord_sf(
    xlim = c(-77.870, -77.850),
    ylim = c(40.795, 40.805),
    expand = FALSE,
    crs = 4326
  ) +
  theme_void() +
  theme(panel.border = element_rect(color = "black", fill = NA,
    linewidth = 2))

# 5. Combine them using Patchwork
final_map <- map_main +
  inset_element(map_inset, left = 0.45, bottom = 0.05,
    right = 0.95, top = 0.45)

final_map

```

Centre County, Pennsylvania



Setup checklist — due by Wednesday

Install R and RStudio

- R: <https://cloud.r-project.org>
- RStudio: <https://posit.co/download/rstudio-desktop>

⚠ Mac users

Install **XQuartz** too — some packages we'll use later in the course need it.

Configure RStudio

- Tools > Global Options... > General
- Restore .RData into workspace at startup: **Unchecked**
- Save workspace to .RData on exit: **Never**

Assignments due Wednesday 5/20

- HW 0: Setup confirmation

Wrap-up & looking ahead

Tomorrow: Real R syntax (variables, functions, vectors)

Due Tuesday before class:

- Read *Data Computing* §2.1–2.8 on Perusall.

Reference links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [Data Computing](#)

Reach out

Stuck on setup? Come to office hours or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu) before Wednesday — don't wait until class.